



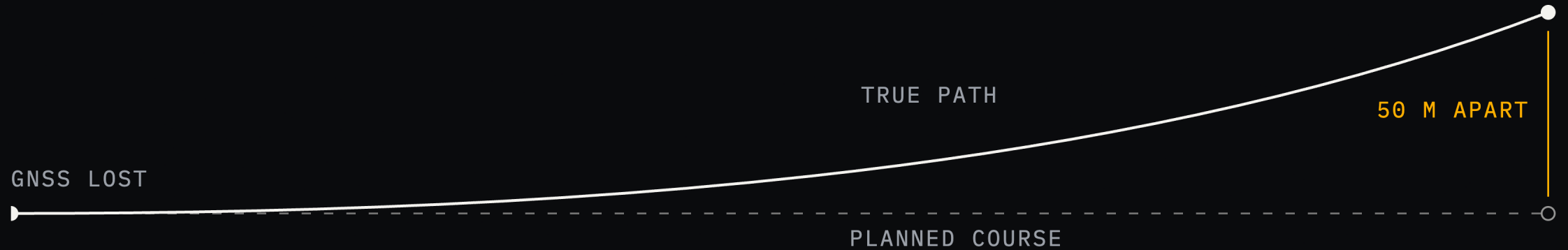
Taipei Drift

We stay on course when GNSS is jammed.

THE PROBLEM



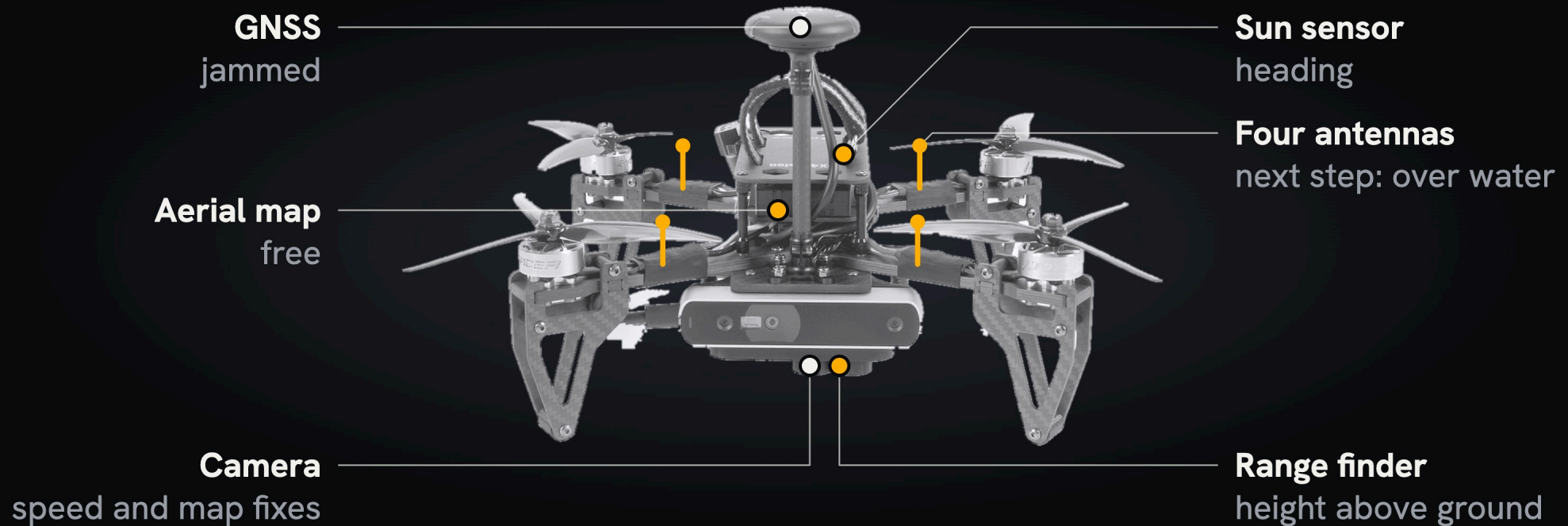
Once GNSS is jammed, a low-cost drone drifts off course.



36 s

Source: own measurement, inertial sensors alone, median of 30 flights of the Mid-Air dataset.

It navigates by what its camera sees.



● added ● already on the drone

TAIPEI DRIFT



Demo

Demo video: drive.google.com/file/d/1NBC1rCQY6rVgWE1i2Bdkl-2W_jYXX9vt/view



The whole drone costs from \$875.

Frontline FPV drone	from \$300
Raspberry Pi 5, 2 GB	\$78
Downward camera	\$29
Memory card, 64 GB	\$32
Sun sensor	from \$10
Laser range finder	\$299
Radio direction finder	from \$127
Together	from \$875
Optional, for absolute position: NVIDIA Jetson Orin Nano Super	\$399



Cheap drones are bought in very large numbers.

48,750

drones in Taiwan's tender

For 2026 and 2027. 34,000 are small multirotors.

4.5 M

FPV drones for Ukraine

Planned purchases in 2025.

\$22.8 B

military drones by 2030

Market forecast.

Proposed: a \$200 licence per drone. Taiwan's small multirotors alone: \$6.8 M.

Sources: CNA, 23 July 2025 (tender notice); Kyiv Independent, March 2025; MarketsandMarkets, May 2025.

NEXT STEPS



Next, and what it takes.

1

A real flight

A small computer, a camera, the range finder, a cheap drone, a test range.

2

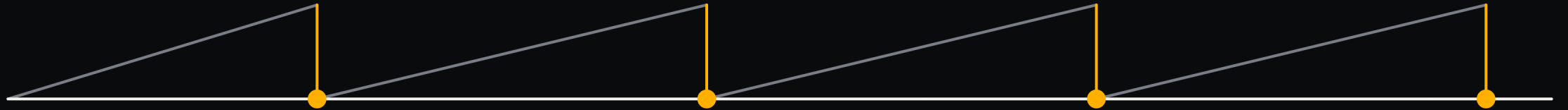
Open water

The radio direction finder, built and measured.

3

Night

A thermal camera against the same map.



**We stay on course
when GNSS is jammed.**



Backup

12 Limits

13 Real photos

14 Each sensor

15 Sun sensor

16 Alternatives

17 Competition

18 Our position

19 Computing



Night, fog and a jammed take-off are still open.

Night	Needs a thermal camera. Not built.
Open water	Ships' radio holds tens of metres. Without it, drift.
Fog, rebuilt ground	The map match fails, and the status says so.
Jammed from take-off	Not covered. It calibrates on a stretch with GNSS.
Range finder	For low flight and hills. Only the demo uses it so far.
What is simulated	Demo and sealed flights. The demo autopilot flies on the true position.

BACKUP / REPLAY OF A REAL FLIGHT, MIAOLI



It holds on real drone photos from Taiwan.

3.3 m

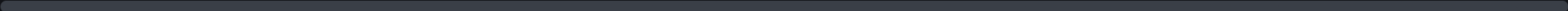
median error over 4 km without GNSS. Dead reckoning alone: 54 m.

Source: own test, docs/research/tuniu-level2-results.md. One DJI flight of 2019, replayed. Barometer simulated. A second matcher of ours.

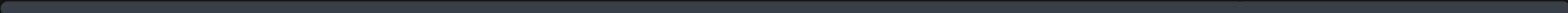


Each sensor we add cuts the error.

3 km inertial sensors alone



2.4 km + sun heading



57 m + camera speed

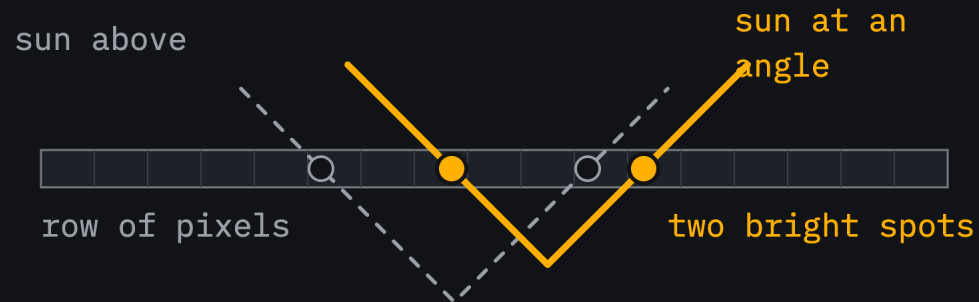


18 m + map fixes



The development flight, used for tuning. On the two sealed flights: 12 m and 32 m.

A slit over one row of pixels reads the sun's direction.



Sunlight through a V-shaped slit falls on the row as **two bright spots**.

Both spots shift together: one angle of the sun. They move closer or apart: the other angle.

Proven	0.1° over $\pm 65^\circ$, 25 readings a second, 35 g, 0.2 W. Flown on six small satellites.
On our drone	With date and time it gives the heading: 1.3° in our simulation. Not built yet.
Limits	Sun below 25° or right overhead: no heading. Under cloud the gyroscope carries on.



Other ways to resist jamming cost far more.

\$20,000 and up: anti-jam antenna (CRPA)

\$20,000 camera navigation with a GPU (Vermeer)

\$15,000 and up: fibre-optic inertial unit

\$1,500 scanning LiDAR (Livox Avia)

\$575 everything we add



The idea is proven. Ours is the low-cost version.

	Taipei Drift	Raptor (Vantor)	VNS01	Vermeer VPS	Eagle Eyes (Ukraine)
Map	Free aerial photos	Licensed 3D data	Satellite imagery	3D terrain	Fresh reconnaissance
Hardware	Camera, sun sensor, range finder, radio, CPU	Camera and a GPU	100 g unit, 5 W	Four cameras, GPU	Not public
Price per drone	\$575 for the kit	Not public	Not public	About \$20,000	Not public
Accuracy	12 to 32 m, simulated	Under 10 m	Up to 30 m	Not stated	Not public
Maturity	Weekend prototype	Flight-tested in Taiwan	Product	Fielded in Ukraine	In combat since 2023

Sources: makers' product pages (Vantor, UAV Navigation), DroneXL May 2026 (Vermeer), The Economist May 2024 (Eagle Eyes), own tests and cost slide.



It is cheap, independent and honestly tested.

Price class	A \$575 kit. The only other public price: about \$20,000.
Independence	Free maps and retail parts. Raptor needs a vendor licence per machine.
Rigour	Code frozen first. Sealed flights run once. Tests declared in advance.



It runs on one CPU core.

11 ms

per camera frame

Room for 90 frames a second.

0.07–2 s

per map fix

One fix every 30 s: under 7 percent of a core.

4 MB

of map per km²

A 64 GB card holds 16,000 km².

Target: a **Raspberry Pi 5, \$78**, 3 to 10 W. Optional: a Jetson Orin Nano Super, \$399, 7 to 25 W.

Sources: own timing on one laptop core, not yet on a drone computer; raspberrypi.com, CNX Software, NVIDIA, Jeff Geerling's board tests.